

# FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

## FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.6: AREA OF A PART OF A CIRCLE

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

Final Explanation: Answer the Farmer's Question. Think about the Laikipia County centre-pivot irrigation farm from our lessons. The driving question is: How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers and artisans can make accurate plans and avoid waste? Answer every section in the space below it. Write each formula you use, show every step of your working, and give units in your final answers (use the  $\pi$  button on your calculator, and give answers to 2 decimal places unless told otherwise). Where you are asked to explain, write full sentences and refer to the farm, the evidence you gathered, or the models you built.

### Part 1: Why the full-circle formula is not enough (Annulus)

**The Laikipia farmer's title deed gives only the area of the full circular plot. (a) Explain why the formula  $A = \pi r^2$  alone cannot tell the farmer the area of the ring-shaped zone between the inner pipe ring and the outer pipe ring. (b) The ring lies between an inner circle of radius 25 m and an outer circle of radius 40 m, both centred on the pivot. Write the formula for the area of an annulus and calculate the area of the ring. (c) A chapati-pan border is also an annulus. Explain in one sentence how the same idea applies to it.**

(a)  $A = \pi r^2$  gives the area of one whole circle only. The ring is not a whole circle: it is the outer circle with the inner circle taken away, so we need to use two radii and subtract. (b) An annulus with outer radius  $R$  and inner radius  $r$  has area  $A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$ . Here  $R = 40$  m and  $r = 25$  m.  $A = \pi(1600 - 625) = \pi \times 975 \approx 3063.05$  m<sup>2</sup>. (c) The chapati-pan border is the area of the whole pan (outer radius) minus the area of the inner flat part (inner radius), so we subtract the smaller circle's area from the larger circle's area in the same way.

### Part 2: The pivot's sweep (Sector, arc length and perimeter)

**The pivot pipe of the maize field is 40 m long, and it has rotated through 120°. (a) What fraction**

(a) A full circle is 360°, so a sector with central angle 120° covers  $120/360 = 1/3$  of the circle. Area and angle are in proportion, so the sector is that same fraction of the whole circle's area. (b)  $A = (\theta/360) \times \pi r^2 = (120/360) \times \pi \times 40^2 = (1/3) \times \pi \times 1600 \approx 1675.52$  m<sup>2</sup>. (c) Arc length  $l = (\theta/360) \times 2\pi r = (1/3) \times 2 \times \pi \times 40 \approx 83.78$  m. Sector perimeter  $P = l + 2r = 83.78 + 2(40) = 163.78$  m. So

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| of the full circle has the pipe swept, and why does that fraction work? (b) Write the formula for the area of a sector and calculate the area of maize watered so far. (c) The farmer wants to fence the edge of this watered sector. Write the formulae for arc length and sector perimeter, and calculate the length of fencing needed. | about 163.78 m of fencing is needed, because the boundary is the curved arc plus the two straight radii. |
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### Part 3: Combining the ring and the wedge (Annular sector)

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| The outer irrigation ring, between radius 25 m and radius 40 m, has been swept through 120°. (a) Explain how the annular sector formula comes from the annulus formula and the sector formula. (b) Calculate the area of this watered wedge-shaped ring. (c) A classmate calculates $(40 - 25)^2$ and then multiplies by $\pi$ and by 120/360. Explain what is wrong with this method. | (a) An annular sector is the large sector minus the small sector with the same angle: $(\theta/360)\pi R^2 - (\theta/360)\pi r^2 = (\theta/360) \times \pi(R^2 - r^2)$ . It joins the annulus idea (subtract the inner area from the outer area) with the sector idea (take the fraction $\theta/360$ ). (b) $A = (120/360) \times \pi \times (40^2 - 25^2) = (1/3) \times \pi \times (1600 - 625) = (1/3) \times \pi \times 975 \approx 1021.02 \text{ m}^2$ . This is one third of the annulus area, which agrees with the angle fraction. (c) The classmate subtracted the radii before squaring. Areas depend on the squares of the radii, so each radius must be squared first and then subtracted: $R^2 - r^2 = 975$ , but $(R - r)^2 = 225$ , which is a different and wrong value. |
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### Part 4: Chord-cut regions and overlapping plots (Segment and common region)

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| (a) A circular Turkana borehole hatch cover has radius 35 cm. A straight locking bar (a chord) cuts off a segment whose central angle is 110°. Write the formula for the area of a segment and calculate the area of this segment. (b) Two circular irrigation plots on the Laikipia farm have radii $r_1 = 42 \text{ m}$ and $r_2 = 35 \text{ m}$ , and their centres are $d = 49 \text{ m}$ apart. Explain the steps you would follow to find the area they | (a) Segment area = sector area – triangle area: $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ . With $r = 35 \text{ cm}$ and $\theta = 110^\circ$ : sector = $(110/360) \times \pi \times 1225 \approx 1175.9 \text{ cm}^2$ ; triangle = $\frac{1}{2} \times 1225 \times \sin 110^\circ \approx 575.6 \text{ cm}^2$ ; segment $\approx 600.3 \text{ cm}^2$ . (b) Draw the chord joining the two intersection points. The lens-shaped common region is made of two segments, one from each circle. For each circle, use the cosine rule in the triangle formed by the two centres and one intersection point to find the half-angle, then double it to get the central angle. Calculate each segment area with the segment formula, then add the two. For the circle of radius 42 m: $\cos\alpha = (r_1^2 + d^2 - r_2^2)/(2r_1d) = (1764 + 2401 - 1225)/(2 \times 42 \times 49) = 2940/4116 \approx 0.7143$ , so $\alpha \approx 44.4^\circ$ and the central angle is $2\alpha \approx 88.8^\circ$ . (c) Shared area $\approx 485.8 + 662.3 \approx 1148 \text{ m}^2$ . It matters because both farmers know exactly how much land is jointly watered, so they can split water costs fairly and avoid counting that land twice in planting and fencing plans. |
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| share, including how the central angles are found. Use the cosine rule to find the central angle at the centre of the circle of radius 42 m. (c) The two segment areas are approximately 485.8 m <sup>2</sup> and 662.3 m <sup>2</sup> . What is the shared area, and why does it matter to the two farmers? |  |
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| Part 5: Answering the farmer's question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| Use everything you have learned to answer the driving question: How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers and artisans can make accurate plans and avoid waste? In your answer: (a) name the five sub-regions of the Laikipia farm we studied and give the formula for each; (b) explain how you decide which formula to use for a shape; (c) give one example, other than the Laikipia farm, where a Kenyan engineer or artisan could use one of these formulae to avoid waste. | (a) Annulus (ring between two pipes): $A = \pi(R^2 - r^2)$ . Sector (wedge swept by the pivot): $A = (\theta/360)\pi r^2$ . Annular sector (wedge-shaped part of a ring): $A = (\theta/360)\pi(R^2 - r^2)$ . Segment (region between a chord and its arc): $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2 \sin \theta$ . Common region of two intersecting circles (lens): the sum of two segment areas, with central angles found using the cosine rule. (b) I look at the structure of the shape: is it a ring, a wedge, a wedge-shaped ring, a region cut by a straight line, or an overlap? Then I break the shape into circle parts, choose the formula that matches each part, and add or subtract their areas. For a ring I subtract the inner circle from the outer; for a sector I take a fraction $\theta/360$ of the circle; for a segment I subtract the triangle from the sector. (c) A road engineer in Kisumu can use the annular sector formula to find the exact area of paint needed for a curved road-marking strip at the Mega roundabout, so no paint is wasted. (Any similar correct Kenyan example is acceptable.) Any part of a circle can be measured exactly by splitting it into these sub-regions and applying the matching formula, so farmers, engineers and artisans can plan seed, water, fencing and materials precisely and avoid waste. |

| RUBRIC                                                         |                                                                                                                                                                |                                                                                                               |                                                                                                                     |
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| Criterion                                                      | Excellent (4)                                                                                                                                                  | Proficient (3)                                                                                                | Developing (1–2)                                                                                                    |
| Selecting and stating the correct formula for each circle part | Names the correct formula for all five sub-regions (annulus, sector, annular sector, segment, common region) and explains why each fits the shape's structure. | States the correct formula for most sub-regions, with only minor slips or limited explanation of why it fits. | States few correct formulae or confuses them, for example omitting the triangle subtraction in the segment formula. |
| Accuracy of calculations and units                             | All calculations are correct, with each radius squared before subtracting, angles used correctly, and answers given in the                                     | Most calculations are correct, with small rounding or unit errors that do not change                          | Several calculation errors, such as subtracting radii before squaring, or units                                     |

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|                                                  | right units (m <sup>2</sup> , cm <sup>2</sup> , m).                                                                                | the method.                                                                                                    | missing or wrong.                                                                                       |
| Structured working and mathematical reasoning    | Working is set out clearly step by step, and the cosine rule steps for the lens are explained and linked to the segments.          | Working is mostly clear, with some steps missing or the reasoning for the common region only partly explained. | Working is hard to follow or answers are given without steps or reasons.                                |
| Connecting to the phenomenon and Kenyan contexts | Clearly links every formula to the Laikipia farm and gives a correct, relevant Kenyan example showing how exact area avoids waste. | Links most answers to the Laikipia farm, and gives a Kenyan example with limited explanation.                  | Little or no link to the farm or Kenyan contexts, and the answer to the driving question is incomplete. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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